

LinearModels:Homework

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```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.0      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.2      v tibble    3.3.1
## v lubridate  1.9.5      v tidyr     1.3.2
## v purrr      1.2.1
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lme4)
```

```
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
```

```
library(emmeans)
```

```
## Welcome to emmeans.
## Caution: You lose important information if you filter this package's results.
## See '? untidy'
```

```
library(multcomp)
```

```
## Loading required package: mvtnorm
## Loading required package: survival
## Loading required package: TH.data
## Loading required package: MASS
##
## Attaching package: 'MASS'
##
## The following object is masked from 'package:dplyr':
```

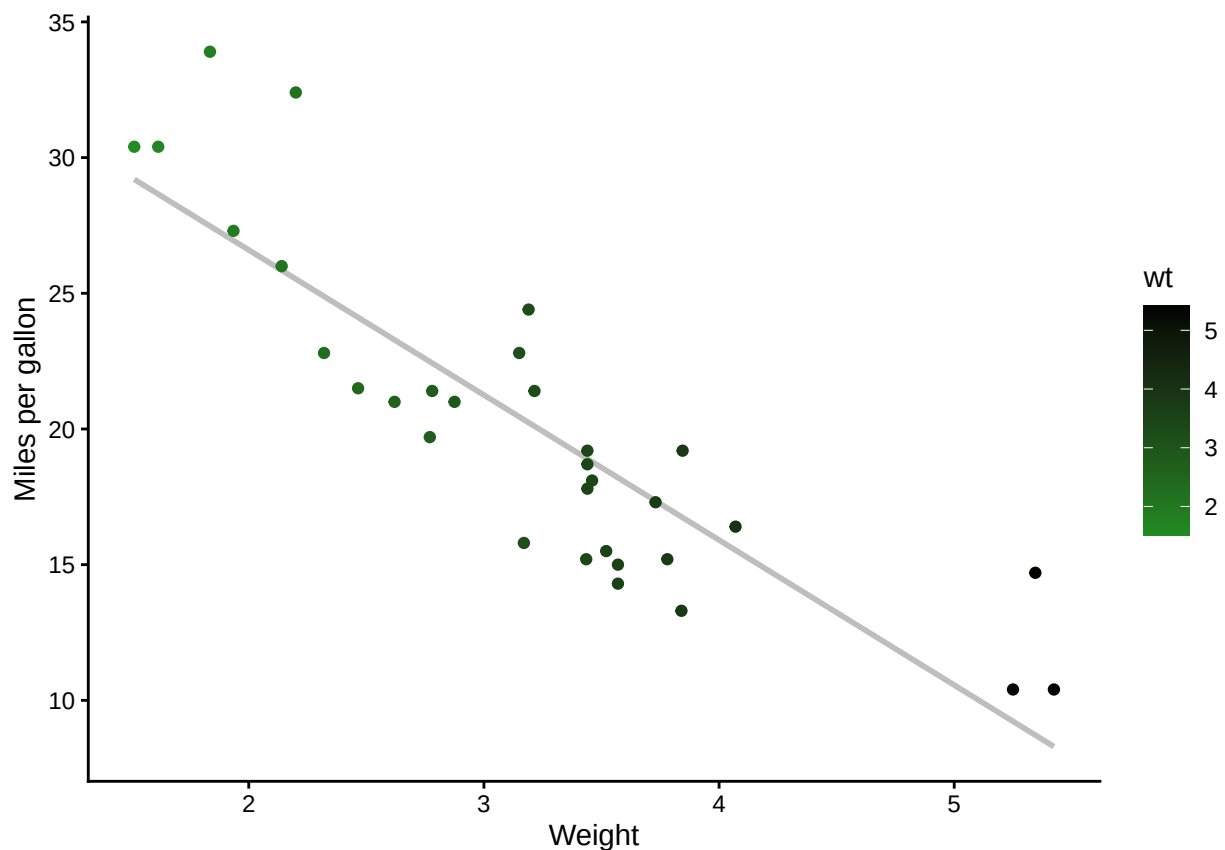
```
##
##   select
##
##
## Attaching package: 'TH.data'
##
## The following object is masked from 'package:MASS':
##
##   geyser
```

```
data("mtcars")
```

```
##Continuous variables
```

```
ggplot(mtcars, aes(x = wt, y = mpg)) +
  geom_smooth(method = lm, se = FALSE, color = "grey") +
  geom_point(aes(color = wt)) +
  xlab("Weight") +
  ylab("Miles per gallon") +
  scale_colour_gradient(low = "forestgreen", high = "black") +
  theme_classic()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



There appears to be a linear relationship between wt and mpg.

```
d1 <- lm(mpg~wt, data = mtcars)
summary(d1)
```

```
##
## Call:
## lm(formula = mpg ~ wt, data = mtcars)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.5432 -2.3647 -0.1252  1.4096  6.8727
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  37.2851     1.8776  19.858 < 2e-16 ***
## wt          -5.3445     0.5591  -9.559 1.29e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.046 on 30 degrees of freedom
## Multiple R-squared:  0.7528, Adjusted R-squared:  0.7446
## F-statistic: 91.38 on 1 and 30 DF,  p-value: 1.294e-10
```

In this case, about 75% variation in y is explained by x.

```
anova(lm(mpg~wt, data = mtcars))
```

```
## Analysis of Variance Table
##
## Response: mpg
##           Df Sum Sq Mean Sq F value    Pr(>F)
## wt           1  847.73   847.73   91.375 1.294e-10 ***
## Residuals    30  278.32     9.28
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Since $p = 1.29e-10$, there is a significant relationship between our x and y variables.

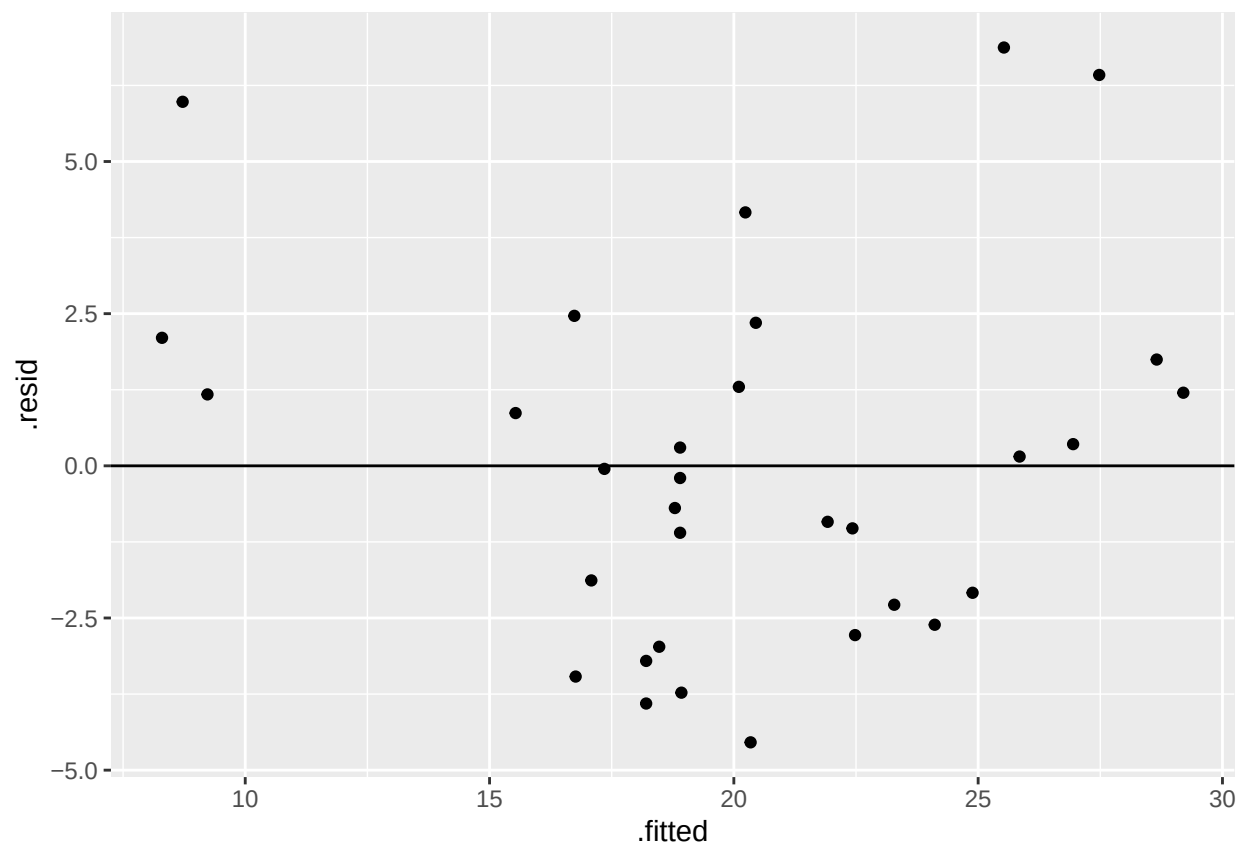
```
cor.test(mtcars$wt, mtcars$mpg)
```

```
##
## Pearson's product-moment correlation
##
## data:  mtcars$wt and mtcars$mpg
## t = -9.559, df = 30, p-value = 1.294e-10
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.9338264 -0.7440872
## sample estimates:
##      cor
## -0.8676594
```

```
model <- lm(mpg~wt, data = mtcars)
```

```
ggplot(model, aes(y = .resid, x = .fitted)) +  
  geom_point() +  
  geom_hline(yintercept = 0)
```

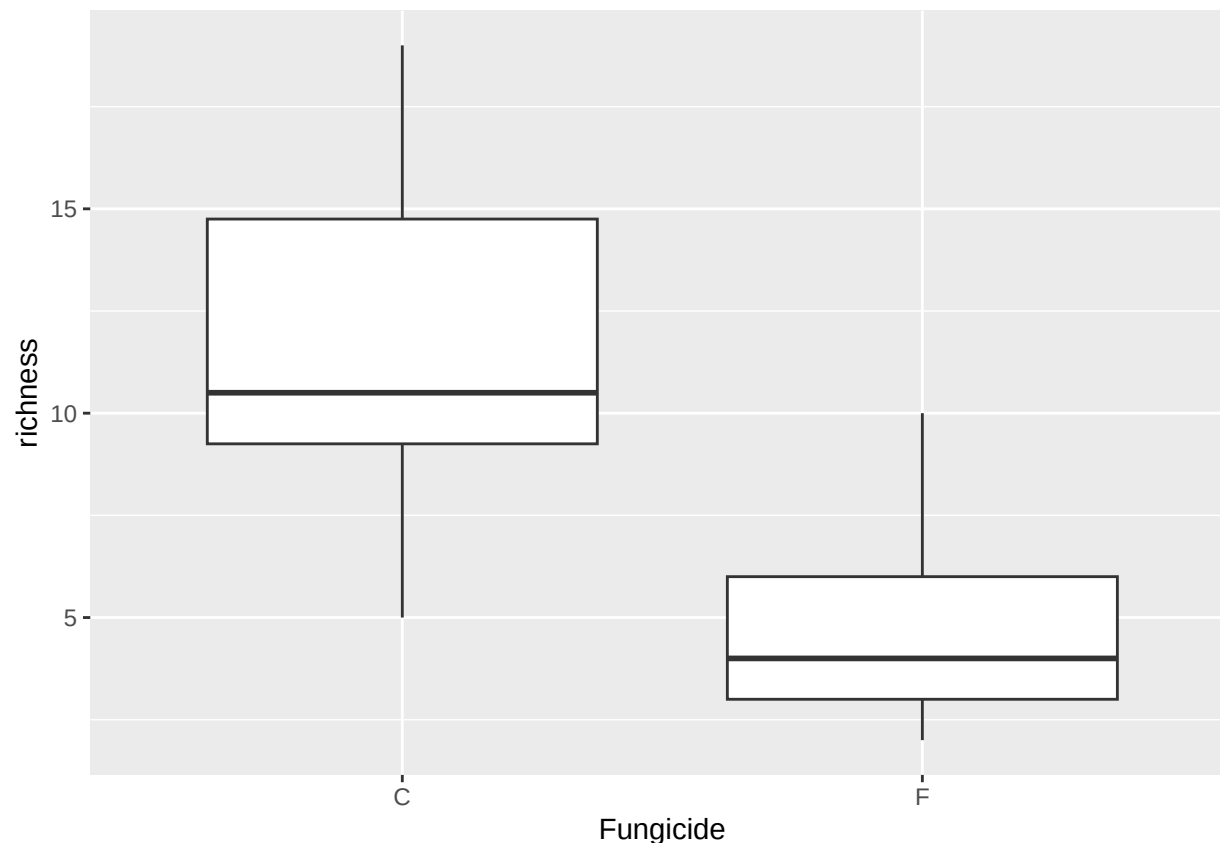
```
## Warning: 'fortify(<lm>)' was deprecated in ggplot2 4.0.0.  
## i Please use 'broom::augment(<lm>)' instead.  
## i The deprecated feature was likely used in the ggplot2 package.  
## Please report the issue at <https://github.com/tidyverse/ggplot2/issues>.  
## This warning is displayed once per session.  
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was  
## generated.
```



```
##Categorical variables
```

```
bull.rich <- read.csv("Data/Bull_richness.csv")
```

```
bull.rich %>%  
  filter(GrowthStage == "V8" & Treatment == "Conv.") %>%  
  ggplot(aes(x = Fungicide, y = richness)) +  
  geom_boxplot()
```



```
bull.rich.sub <- bull.rich %>%
  filter(GrowthStage == "V8" & Treatment == "Conv.")
t.test(richness~Fungicide, data = bull.rich.sub)
```

```
##
## Welch Two Sample t-test
##
## data: richness by Fungicide
## t = 4.8759, df = 17.166, p-value = 0.0001384
## alternative hypothesis: true difference in means between group C and group F is not equal to 0
## 95 percent confidence interval:
##  4.067909 10.265425
## sample estimates:
## mean in group C mean in group F
##      11.750000      4.583333
```

```
t.test(richness~Fungicide, data = bull.rich.sub, var.equal = TRUE)
```

```
##
## Two Sample t-test
##
## data: richness by Fungicide
## t = 4.8759, df = 22, p-value = 7.118e-05
## alternative hypothesis: true difference in means between group C and group F is not equal to 0
```

```
## 95 percent confidence interval:
##    4.118459 10.214875
## sample estimates:
## mean in group C mean in group F
##      11.750000      4.583333
```

```
summary(lm(richness~Fungicide, data = bull.rich.sub))
```

```
##
## Call:
## lm(formula = richness ~ Fungicide, data = bull.rich.sub)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.7500 -1.7500 -0.6667  2.2500  7.2500
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    11.750      1.039   11.306 1.24e-10 ***
## FungicideF      -7.167      1.470   -4.876 7.12e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.6 on 22 degrees of freedom
## Multiple R-squared:  0.5194, Adjusted R-squared:  0.4975
## F-statistic: 23.77 on 1 and 22 DF,  p-value: 7.118e-05
```

```
anova(lm(richness~Fungicide, data = bull.rich.sub))
```

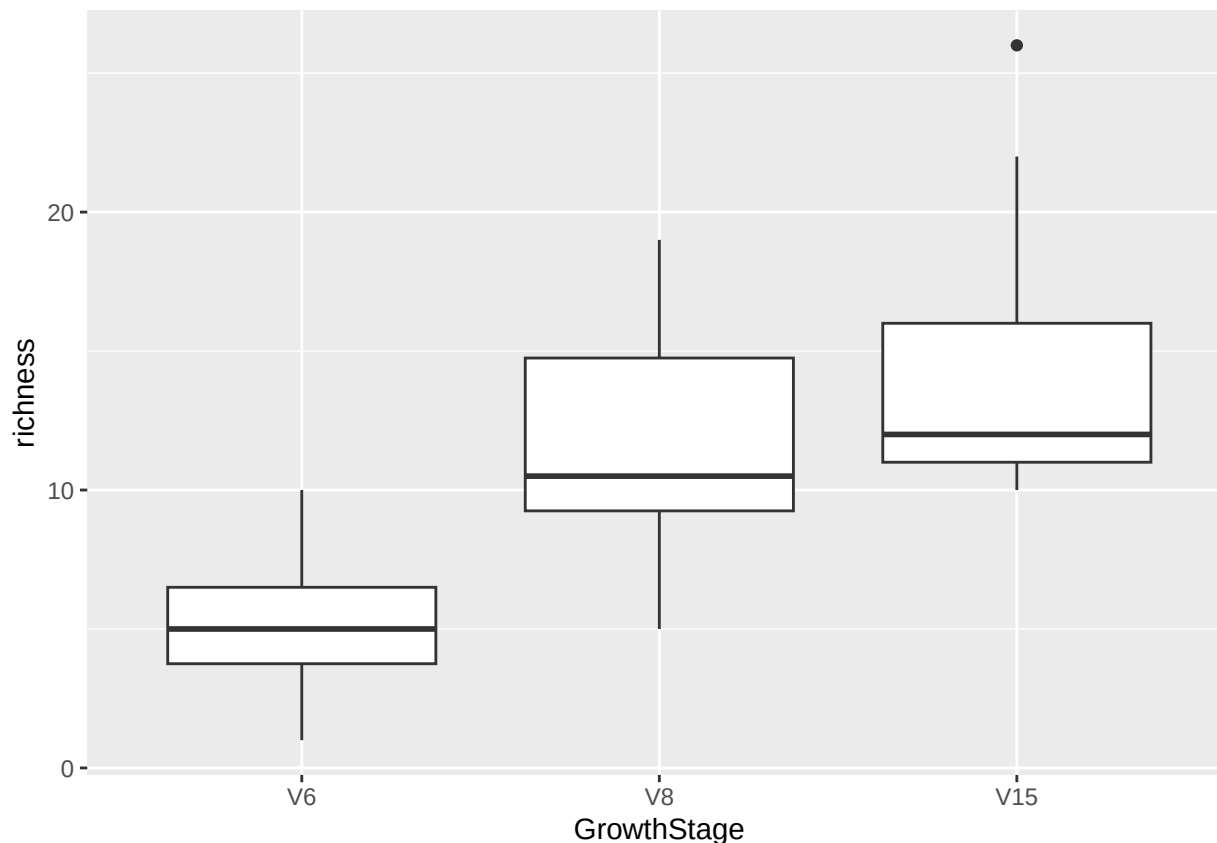
```
## Analysis of Variance Table
##
## Response: richness
##           Df Sum Sq Mean Sq F value    Pr(>F)
## Fungicide  1 308.17  308.167   23.774 7.118e-05 ***
## Residuals 22 285.17   12.962
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Assuming equal variance in groups and performing a two-sample t-test is the same result as a linear model and ANOVA on the linear model. The p-value tells us that the slope of the line is different than 0.

##ANOVAS- Continuous Y and multiple Categorical X.

```
bull.rich.sub2 <- bull.rich %>%
  filter(Fungicide == "C" & Treatment == "Conv." & Crop == "Corn")
```

```
bull.rich.sub2$GrowthStage <- factor(bull.rich.sub2$GrowthStage, levels = c("V6", "V8", "V15"))
ggplot(bull.rich.sub2, aes(x = GrowthStage, y = richness)) +
  geom_boxplot()
```



```
lm.growth <- lm(richness ~ GrowthStage, data = bull.rich.sub2)
summary(lm.growth)
```

```
##
## Call:
## lm(formula = richness ~ GrowthStage, data = bull.rich.sub2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.750 -2.625 -1.000  2.250 11.583
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      5.250      1.208   4.348 0.000124 ***
## GrowthStageV8      6.500      1.708   3.806 0.000581 ***
## GrowthStageV15     9.167      1.708   5.368 6.23e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.183 on 33 degrees of freedom
## Multiple R-squared:  0.4803, Adjusted R-squared:  0.4488
## F-statistic: 15.25 on 2 and 33 DF, p-value: 2.044e-05
```

```
anova(lm.growth)
```

```
## Analysis of Variance Table
##
## Response: richness
##           Df Sum Sq Mean Sq F value    Pr(>F)
## GrowthStage  2 533.56  266.778   15.247 2.044e-05 ***
## Residuals   33 577.42   17.497
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(aov(richness ~ GrowthStage, data = bull.rich.sub2))
```

```
##           Df Sum Sq Mean Sq F value    Pr(>F)
## GrowthStage  2  533.6   266.8   15.25 2.04e-05 ***
## Residuals   33  577.4    17.5
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
lsmeans <- emmeans(lm.growth, ~GrowthStage) # estimate lsmeans of variety within siteXyear
Results_lsmeans <- cld(lsmeans, alpha = 0.05, reversed = TRUE, details = TRUE) # contrast with Tukey adjustment
Results_lsmeans
```

```
## $emmeans
## GrowthStage emmean    SE df lower.CL upper.CL .group
## V15          14.42 1.21 33    11.96    16.87    1
## V8            11.75 1.21 33     9.29    14.21    1
## V6             5.25 1.21 33     2.79     7.71    2
##
## Confidence level used: 0.95
## P value adjustment: tukey method for comparing a family of 3 estimates
## significance level used: alpha = 0.05
## NOTE: If two or more means share the same grouping symbol,
##       then we cannot show them to be different.
##       But we also did not show them to be the same.
##
## $comparisons
## contrast estimate    SE df t.ratio p.value
## V8 - V6         6.50 1.71 33    3.806 0.0016
## V15 - V6         9.17 1.71 33    5.368 <0.0001
## V15 - V8         2.67 1.71 33    1.562 0.2763
##
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
##Understanding Interactions
```

```
bull.rich.sub3 <- bull.rich %>%
  filter(Treatment == "Conv." & Crop == "Corn")

bull.rich.sub3$GrowthStage <- factor(bull.rich.sub3$GrowthStage, levels = c("V6", "V8", "V15"))

# write it like this
lm.inter <- lm(richness ~ GrowthStage + Fungicide + GrowthStage:Fungicide, data = bull.rich.sub3)
```


or like this

```
lm(richness ~ GrowthStage*Fungicide, data = bull.rich.sub3)
```

```
##
## Call:
## lm(formula = richness ~ GrowthStage * Fungicide, data = bull.rich.sub3)
##
## Coefficients:
##              (Intercept)              GrowthStageV8
##                   5.2500                   6.5000
##           GrowthStageV15              FungicideF
##                   9.1667                   -1.2500
## GrowthStageV8:FungicideF GrowthStageV15:FungicideF
##                   -5.9167                   0.3333
```

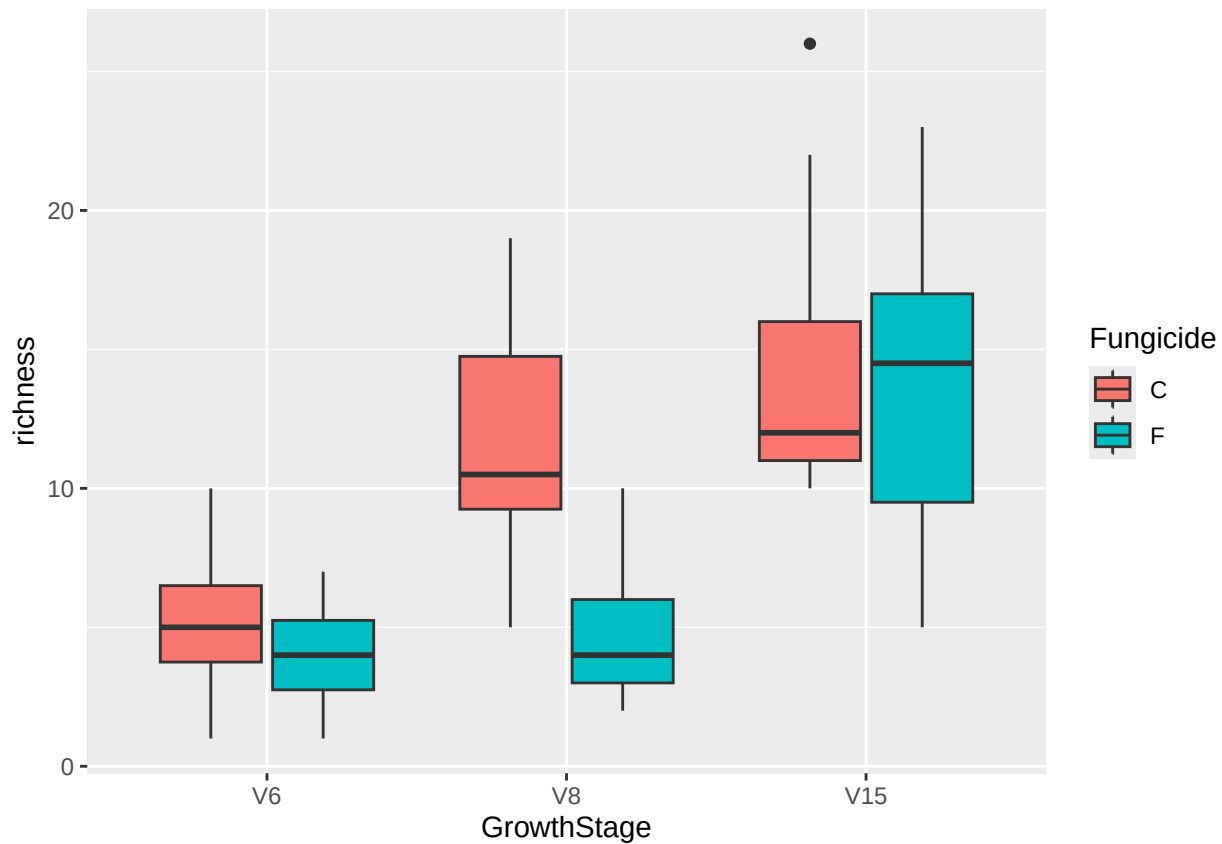
```
summary(lm.inter) # significant terms
```

```
##
## Call:
## lm(formula = richness ~ GrowthStage + Fungicide + GrowthStage:Fungicide,
##     data = bull.rich.sub3)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8.5000 -2.4167 -0.4167  2.0625 11.5833
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      5.2500     1.1029   4.760 1.10e-05 ***
## GrowthStageV8      6.5000     1.5597   4.167 9.15e-05 ***
## GrowthStageV15     9.1667     1.5597   5.877 1.51e-07 ***
## FungicideF       -1.2500     1.5597  -0.801  0.42575
## GrowthStageV8:FungicideF -5.9167     2.2057  -2.682  0.00923 **
## GrowthStageV15:FungicideF  0.3333     2.2057   0.151  0.88034
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.82 on 66 degrees of freedom
## Multiple R-squared:  0.5903, Adjusted R-squared:  0.5593
## F-statistic: 19.02 on 5 and 66 DF,  p-value: 1.144e-11
```

```
anova(lm.inter) # The interaction term is signifant.
```

```
## Analysis of Variance Table
##
## Response: richness
##              Df Sum Sq Mean Sq F value    Pr(>F)
## GrowthStage    2 1065.58   532.79  36.5027 2.113e-11 ***
## Fungicide       1  174.22   174.22  11.9363 0.0009668 ***
## GrowthStage:Fungicide 2  148.36    74.18   5.0823 0.0088534 **
## Residuals     66  963.33    14.60
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
bull.rich.sub3 %>%
  ggplot(aes(x = GrowthStage, y = richness, fill = Fungicide)) +
  geom_boxplot()
```



```
lsmeans <- emmeans(lm.inter, ~Fungicide|GrowthStage) # estimate lsmeans of variety within siteXyear
Results_lsmeans <- cld(lsmeans, alpha = 0.05, reversed = TRUE, details = TRUE) # contrast with Tukey aj
Results_lsmeans
```

```
## $emmeans
## GrowthStage = V6:
##   Fungicide emmean SE df lower.CL upper.CL .group
##   C         5.25 1.1 66    3.05    7.45    1
##   F         4.00 1.1 66    1.80    6.20    1
##
## GrowthStage = V8:
##   Fungicide emmean SE df lower.CL upper.CL .group
##   C        11.75 1.1 66    9.55   13.95    1
##   F         4.58 1.1 66    2.38    6.79    2
##
## GrowthStage = V15:
##   Fungicide emmean SE df lower.CL upper.CL .group
##   C        14.42 1.1 66   12.21   16.62    1
##   F        13.50 1.1 66   11.30   15.70    1
##
## Confidence level used: 0.95
```

```
## significance level used: alpha = 0.05
## NOTE: If two or more means share the same grouping symbol,
##       then we cannot show them to be different.
##       But we also did not show them to be the same.
##
```

```
## $comparisons
## GrowthStage = V6:
## contrast estimate SE df t.ratio p.value
## C - F           1.250 1.56 66  0.801  0.4258
##
## GrowthStage = V8:
## contrast estimate SE df t.ratio p.value
## C - F           7.167 1.56 66  4.595 <0.0001
##
## GrowthStage = V15:
## contrast estimate SE df t.ratio p.value
## C - F           0.917 1.56 66  0.588  0.5587
```

```
##Mixed Effect models
```

```
lme0 <- lm(richness ~ GrowthStage*Fungicide, data = bull.rich.sub3)
lme1 <- lmer(richness ~ GrowthStage*Fungicide + (1|Rep), data = bull.rich.sub3)
summary(lme1)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: richness ~ GrowthStage * Fungicide + (1 | Rep)
## Data: bull.rich.sub3
##
## REML criterion at convergence: 378.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.4664 -0.5966 -0.1788  0.6257  2.9101
##
## Random effects:
## Groups Name Variance Std.Dev.
## Rep (Intercept) 0.7855 0.8863
## Residual 13.9533 3.7354
## Number of obs: 72, groups: Rep, 4
##
## Fixed effects:
## Estimate Std. Error t value
## (Intercept) 5.2500 1.1658 4.503
## GrowthStageV8 6.5000 1.5250 4.262
## GrowthStageV15 9.1667 1.5250 6.011
## FungicideF -1.2500 1.5250 -0.820
## GrowthStageV8:FungicideF -5.9167 2.1566 -2.743
## GrowthStageV15:FungicideF 0.3333 2.1566 0.155
##
## Correlation of Fixed Effects:
## (Intr) GrwSV8 GrSV15 FngcdF GSV8:F
## GrowthStgV8 -0.654
## GrwthStgV15 -0.654 0.500
```

```
## FungicideF -0.654 0.500 0.500
## GrwthSV8:FF 0.462 -0.707 -0.354 -0.707
## GrwtSV15:FF 0.462 -0.354 -0.707 -0.707 0.500
```

```
summary(lme0)
```

```
##
## Call:
## lm(formula = richness ~ GrowthStage * Fungicide, data = bull.rich.sub3)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8.5000 -2.4167 -0.4167  2.0625 11.5833
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      5.2500     1.1029   4.760 1.10e-05 ***
## GrowthStageV8      6.5000     1.5597   4.167 9.15e-05 ***
## GrowthStageV15     9.1667     1.5597   5.877 1.51e-07 ***
## FungicideF      -1.2500     1.5597  -0.801  0.42575
## GrowthStageV8:FungicideF -5.9167     2.2057  -2.682  0.00923 **
## GrowthStageV15:FungicideF  0.3333     2.2057   0.151  0.88034
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.82 on 66 degrees of freedom
## Multiple R-squared:  0.5903, Adjusted R-squared:  0.5593
## F-statistic: 19.02 on 5 and 66 DF, p-value: 1.144e-11
```

```
summary(lme1)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: richness ~ GrowthStage * Fungicide + (1 | Rep)
## Data: bull.rich.sub3
##
## REML criterion at convergence: 378.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.4664 -0.5966 -0.1788  0.6257  2.9101
##
## Random effects:
## Groups Name Variance Std.Dev.
## Rep (Intercept) 0.7855 0.8863
## Residual 13.9533 3.7354
## Number of obs: 72, groups: Rep, 4
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept)      5.2500     1.1658   4.503
## GrowthStageV8      6.5000     1.5250   4.262
## GrowthStageV15     9.1667     1.5250   6.011
## FungicideF      -1.2500     1.5250  -0.820
```

```

## GrowthStageV8:FungicideF   -5.9167      2.1566   -2.743
## GrowthStageV15:FungicideF   0.3333      2.1566    0.155
##
## Correlation of Fixed Effects:
##      (Intr) GrwSV8 GrSV15 FngcdF GSV8:F
## GrowthStgV8 -0.654
## GrwthStgV15 -0.654  0.500
## FungicideF  -0.654  0.500  0.500
## GrwthSV8:FF  0.462 -0.707 -0.354 -0.707
## GrwtSV15:FF  0.462 -0.354 -0.707 -0.707  0.500

```